



Stockholm, May 2, 2026

To the Academy of Finland

Re: Letter of Collaboration in support of the application by Maarit Korpi-Lagg

Dear Members of the Review Committee,

I am writing to express my strong support for the funding application submitted by Maarit Korpi-Lagg and Frederick Gent concerning their proposed research on dust modelling. I am highly enthusiastic about this project, which represents a natural and timely continuation and significant expansion of a research programme that I initiated over a decade ago. Should the Academy of Finland decide to fund this proposal, I will be fully committed to contributing as an active collaborator.

Below, I outline the nature of my involvement in accordance with the Academy's guidelines.

Collaborator Information

- **Name:** Lars Mattsson
- **Position:** Assistant Professor in Astrophysics
- **Affiliation:** Nordita, Stockholm University

Purpose, Objectives, and Added Value of the Collaboration

The primary aim of my participation is to contribute the expertise and experience I have developed over the past decade as principal investigator of the original project (initiated in 2016) and its subsequent extension (from 2022 onward). My research has focused on computational astrophysics, with particular emphasis on dust evolution and modelling using the *Pencil Code*, which remains my principal research tool.

This project builds directly on that foundation, and I see substantial potential for advancing the state of the art through the integration of new physical processes, improved numerical treatments, and closer connections to observational constraints. The proposed work is well positioned to deliver significant progress in our understanding of dust physics in astrophysical environments.

In practical terms, my collaboration will include:



- **Scientific guidance and mentorship:** Regular virtual meetings with the principal investigators and the prospective PhD student to support model development, numerical implementation, and the integration of simulation results with observational or theoretical frameworks.
- **Research visits and exchange:** Active participation in reciprocal visits, including hosting collaborators at Nordita and visiting the host institution, to facilitate intensive collaboration, training, and knowledge transfer.
- **Joint analysis and dissemination:** Close involvement in the interpretation of results, co-authorship of scientific publications, and contribution to the broader dissemination of outcomes within the international research community.

Through these activities, I will contribute both technical expertise and strategic input, ensuring continuity with prior developments while supporting the project's ambition to move beyond the current state of the field. Of critical relevance to this project is the access to Lumi highly parallel GPU infrastructure. I will contribute to the extension of the dust processing modules of Pencil Code updated with Paperboats functionality to the GPU based PC-A code.

Overall Assessment

This project is both timely and scientifically compelling. It addresses key open questions in computational astrophysics using a robust and well-motivated methodological framework. In my assessment, it represents a clear step forward in the field and has strong potential for high-impact scientific outcomes.

I therefore wholeheartedly support this application and look forward to collaborating closely with the team to further develop and extend this line of research.

Please do not hesitate to contact me should you require any additional information regarding my planned contribution.

Yours sincerely,

Lars Mattsson
Assistant Professor in Astrophysics